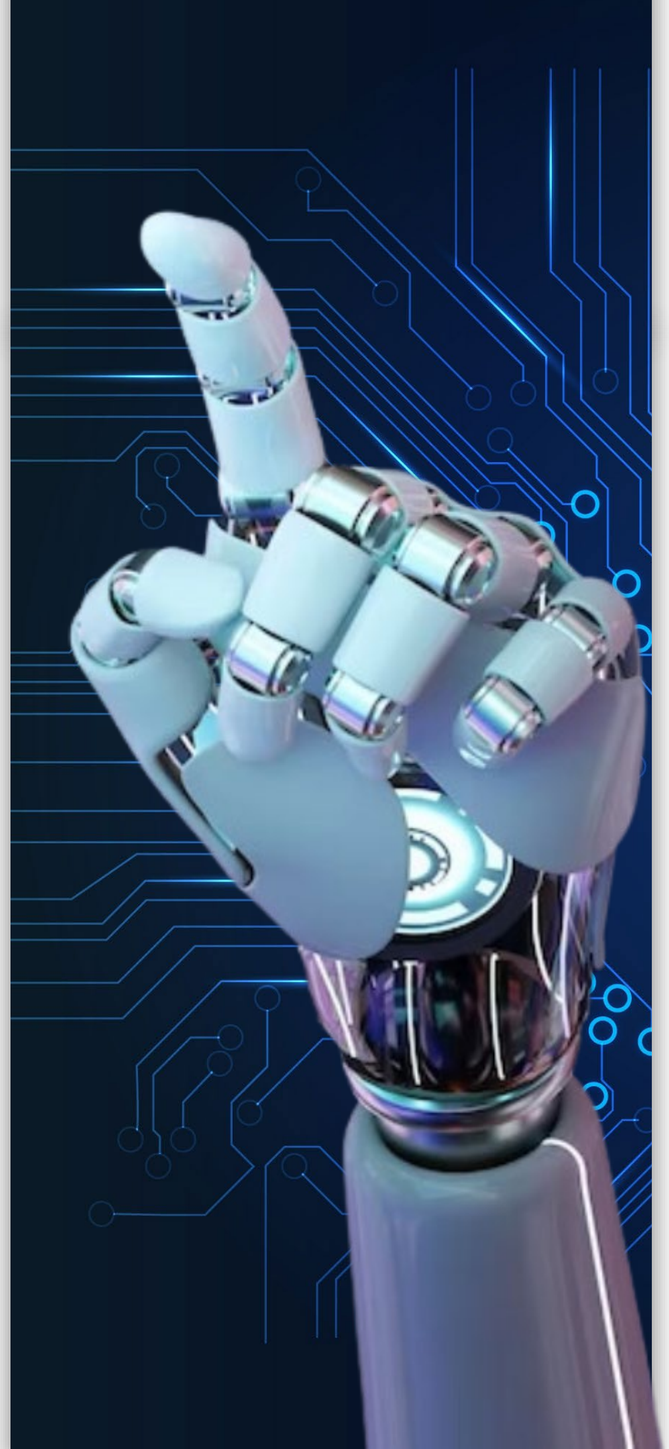


REAL WORLD DATA PLATFORMS

Jeffrey Fine, MPH

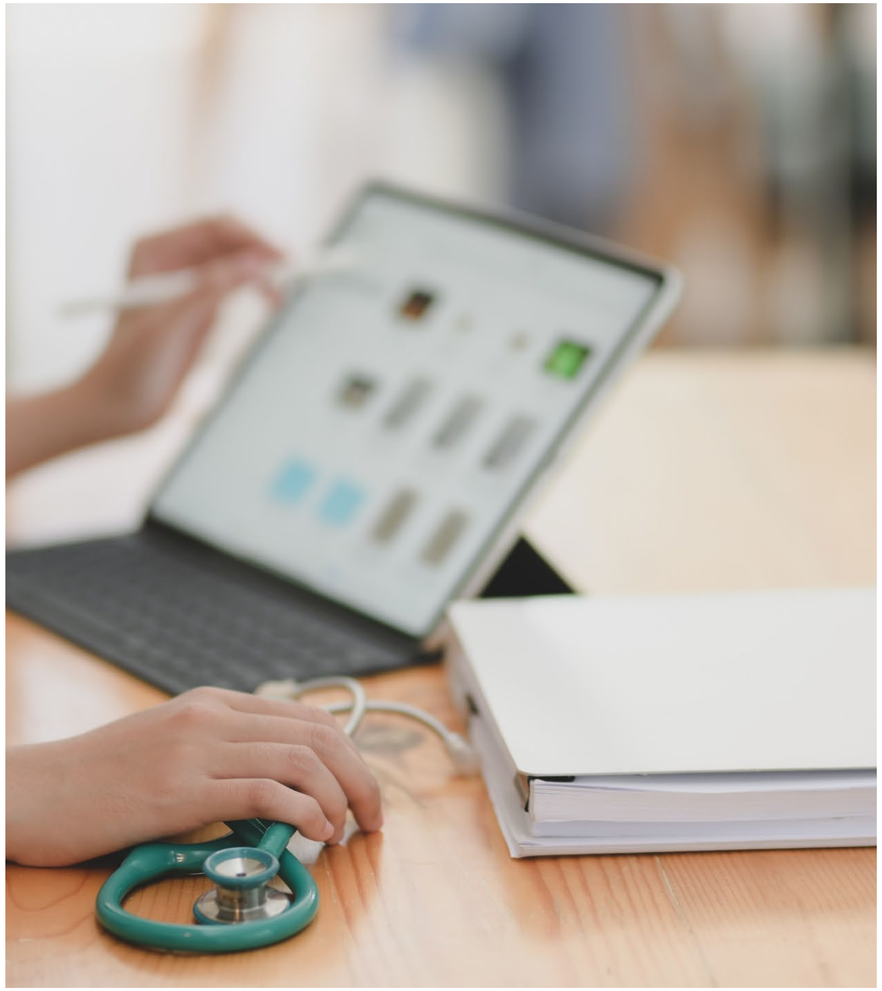
May 28, 2026

UCDAVIS
UNIVERSITY OF CALIFORNIA

Agenda

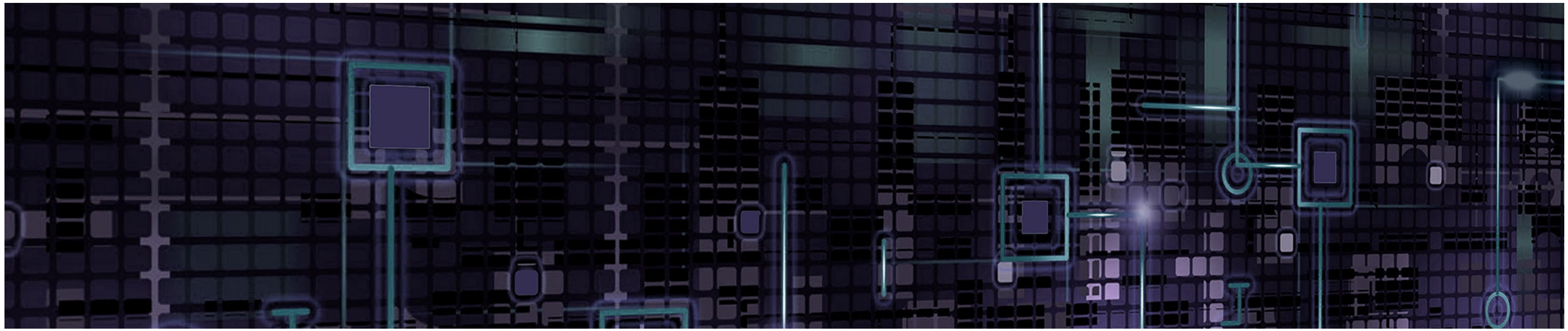
1. What are Real World Data Platforms and What Do They Offer?
2. Manuscripts using this Data
3. Concerns about this data using a Case Study
4. How Should We Advice

REAL WORLD DATA PLATFORMS



- Epic COSMOS
- TriNetX
- National COVID Cohort Collaborative
- Evolve to Next-Gen ACT
- NIH All of Us

ADVANTAGES



- Accessibility beca use institutions invest to make the resource free to researchers (university pays for it).
- Up-to-date data with investigations relevant to current clinical contexts.
- Large sample size enabling research on rare diseases and subgroup analyses.
- Generalizability beyond strict clinical trial criteria.

WHAT IS TRINETX

“Since TriNetX’s inception in 2013, TriNetX supported life science companies and healthcare systems around the world, empowering researchers to find answers to their clinical questions and drive scientific discovery to improve patient lives.”

- Data is sourced directly from 11,000+ healthcare provider sites across more than 20 countries.
- Data is kept within those institutions to protect accuracy, privacy, and trust.
- TriNetX combines this data with scientific expertise and technology to facilitate research design and data analysis for their clients.



TriNetX Continued

01.

Cohort building

02.

Search and select required codes and clinical concepts

03.

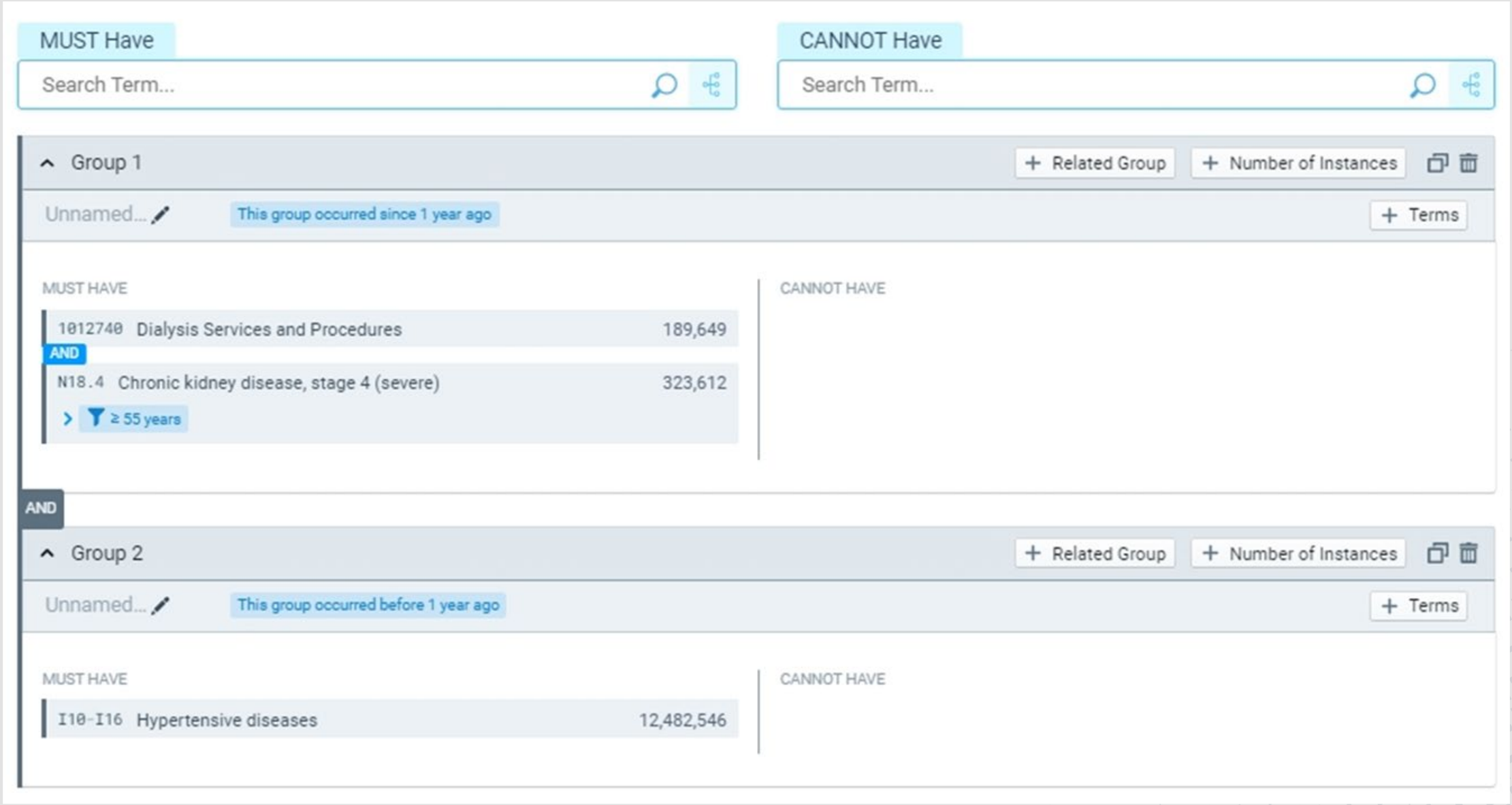
Apply criteria:

- Temporal Logic
- Exclusions
- Inclusions

04.

•Rate of Arrival – predicts volume of new patients who meet your criteria in the next year

FILTERING DATA



SUMMARY STATISTICS OF COHORT

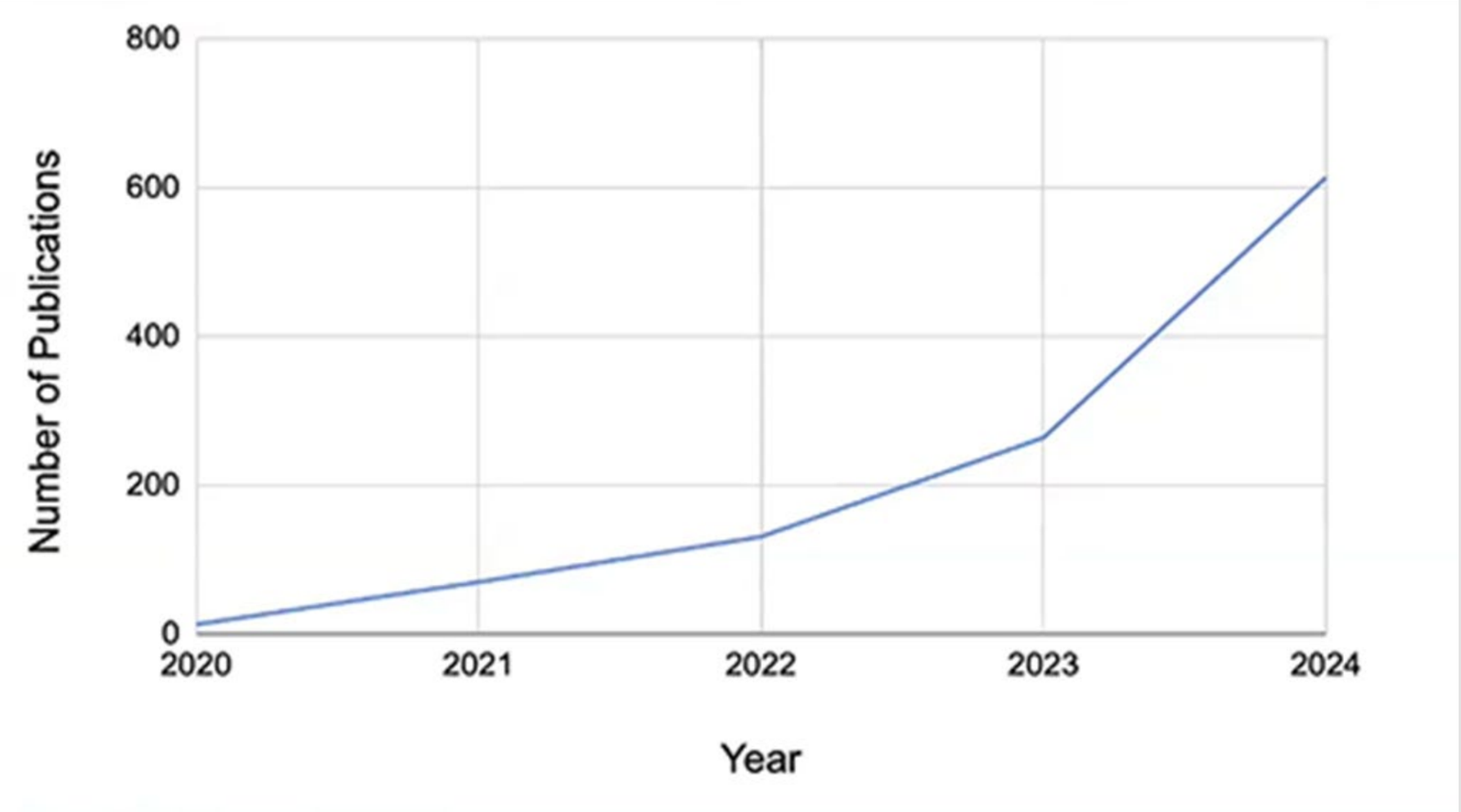
Metabolic Panel		Mean ± SD	Min	Max	Patients	% of Cohort
9029	Sodium [Moles/volume] in Serum, Plasma or Blood	139 ± 4.17	90	200	159,306	88%
9028	Potassium [Moles/volume] in Serum, Plasma or Blood	4.38 ± 0.682	0	25	158,218	87%
9023	Chloride [Moles/volume] in Serum, Plasma or Blood	103 ± 6.38	0	199	157,563	87%
9021	Bicarbonate [Moles/volume] in Serum, Plasma or Blood	24.5 ± 4.68	0	100	156,107	86%
9030	Urea nitrogen [Mass/volume] in Serum, Plasma or Blood	43 ± 25.5	1	390	148,478	82%
9024	Creatinine [Mass/volume] in Serum, Plasma or Blood	2.94 ± 3.04	0	200	162,247	89%
9025	Glucose [Mass/volume] in Serum, Plasma or Blood	141 ± 70.2	0	1,549	159,660	88%
9022	Calcium [Mass/volume] in Serum, Plasma or Blood	8.98 ± 0.856	0	20	158,631	87%
9026	Magnesium [Mass/volume] in Serum, Plasma or Blood	2.03 ± 0.43	0	16	108,328	60%
9027	Phosphate [Mass/volume] in Serum, Plasma or Blood	4 ± 1.38	0	55.3	132,927	73%

ROW LEVEL DATA

patient_id	diagnosis_date	observation_date	tumor_site_code	morphology_code
66e136	20161125	20161125	C61	8140/3
920d5b	20120606	20120606	C50.1	8500/3
920d5b	20120606	20120606	C50.1	8500/3
920d5b	20120606	20120606	C50.1	8500/3
355af3	20180105	20180105	C18.0	8480/3
0cbede	20170607	20170607	C61	8140/3
1dcd8d	20160925	20160925	C61	8140/3
1dcd8d	20160925	20160925	C61	8140/3
46b41f	20160901	20160901	C50.8	8500/3
46b41f	20160901	20160901	C50.8	8500/3
46b41f	20160901	20160901	C50.8	8500/3
98ad3f	20081108	20081108	C50.8	8500/3
739863	20160927	20160927	C61	8140/3

EASE OF USE

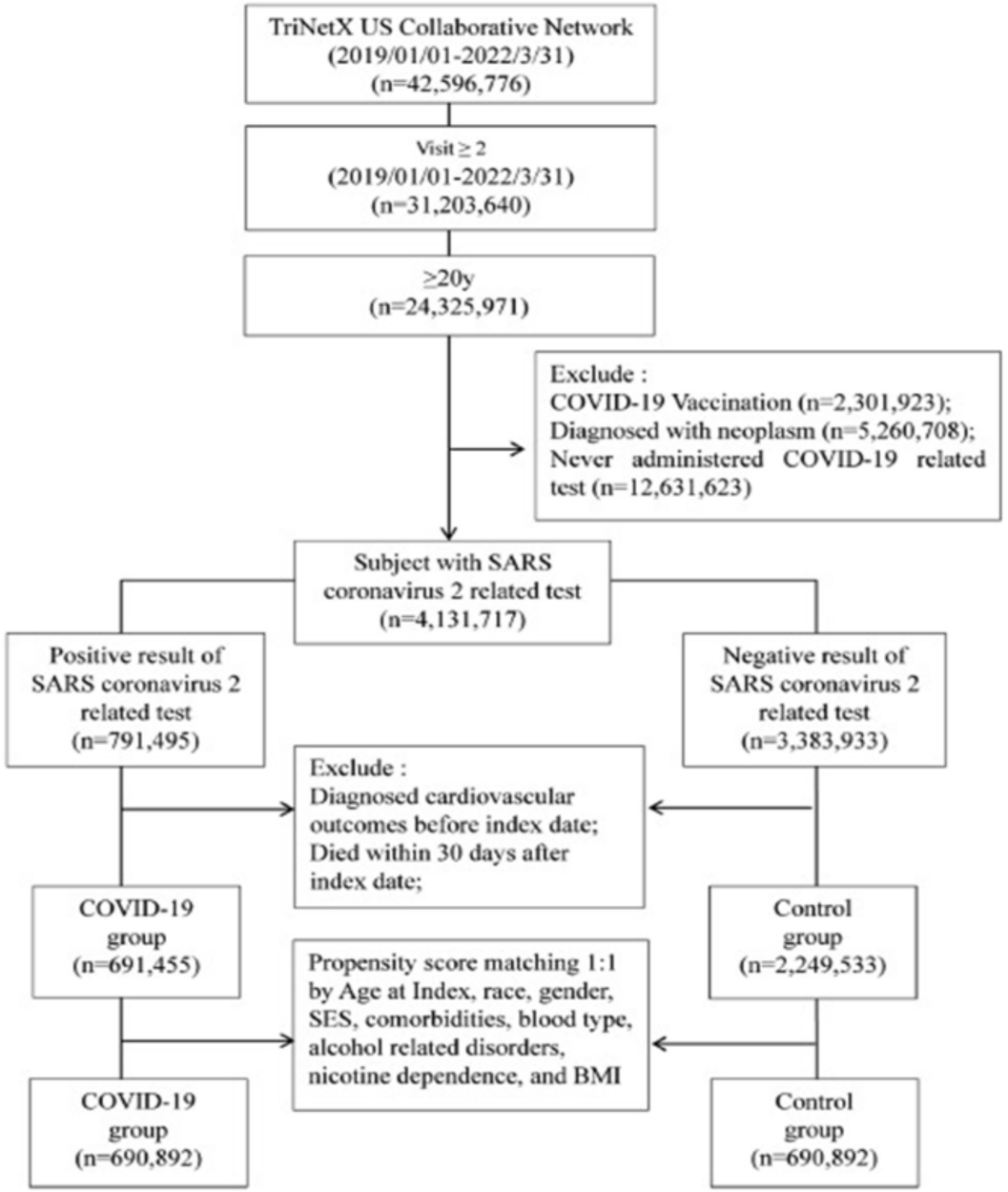
TriNetX analytic tool kit is easily accessible



A Pubmed search shows an increase in the number of papers published using TriNetX

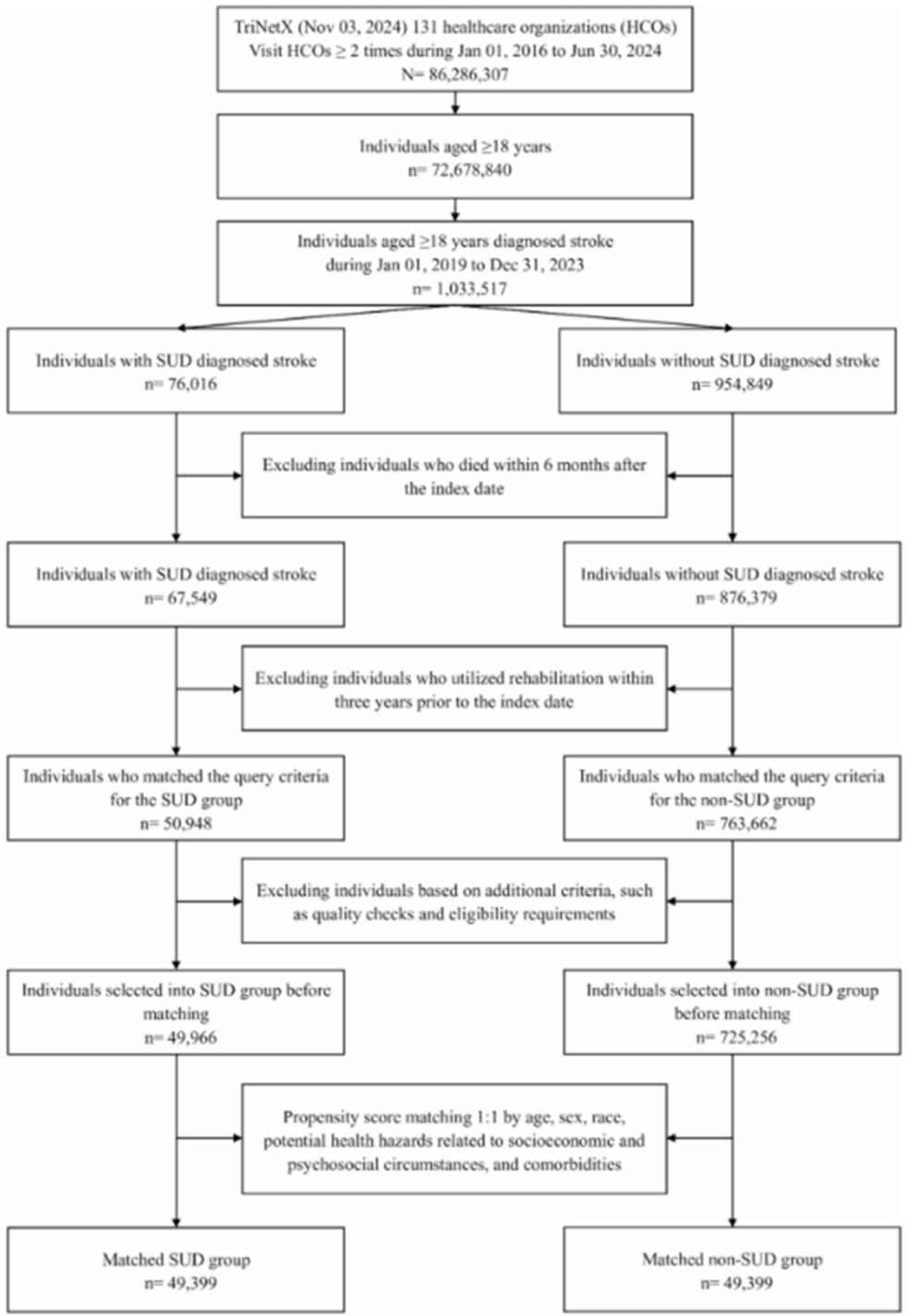
High Impact Manuscripts

- Long-term cardiovascular outcomes in COVID-19 survivors among non-vaccinated population: A retrospective cohort study from the TriNetX US collaborative networks
  - Wang W, Wang C, Wang S et al. *EClinicalMedicine* 53 (2022).



High Impact Manuscripts

- Reduced rehabilitation continuity after stroke in patients with substance use disorder based on a TriNetX retrospective cohort study
  - Kao, Hsiao-Han, et al. *Scientific Reports* (2026).



High Impact Manuscripts

ESKETAMINE COMBINED WITH SSRI OR SNRI FOR TREATMENT-RESISTANT DEPRESSION

- Retrospective Cohort Study of Adults with TRD looking at differences in all-cause mortality, hospitalization, depression relapse, and suicide attempts between patient's treated with esketamine combined with SSRIs or an esketamine combined with SNRI.
- Kaplan-Meier survival analysis was used
- Risk ratios and Odds ratios were also calculated
- Once again, propensity score matching was done
- Results showed patients with esketamine combined with SNRI had lower mortality, hospitalization rates, and depressive relapse
    - Patients with esketamine with SSRI had fewer suicide attempts

ESKETAMINE+SSRI VS. ESKETAMINE+SN

Table 2. Between-Group Comparisons of the Study Variables

Table 2. Between-Group Comparisons of the Study Variables

Variable in study (esketamine + SSRI vs esketamine + SNRI)	Risk difference (95% CI)	*z*	*P* value	Risk ratio (95% CI)	Odds ratio (95% CI)
Mortality	0.038 (0.034 to 0.042)	17.36	<.001	1.720 (1.616 to 1.830)	1.792 (1.676 to 1.915)
Hospitalization	0.001 (0.000 to 0.002)	3.754	<.001	3.009 (1.644 to 5.508)	3.012 (1.644 to 5.516)
Depression relapse	0.063 (0.055 to 0.072)	15.137	<.001	1.427 (1.363 to 1.495)	1.542 (1.458 to 1.632)
Suicide attempts	−0.001 (−0.003 to 0.000)	−2.067	.04	0.701 (0.500 to 0.984)	0.700 (0.499 to 0.983)

Abbreviations: SNRI, serotonin-norepinephrine reuptake inhibitor; SSRI, serotonin reuptake inhibitor.

QUESTIONABLE Reliability in ESKETAMINE STUDY

- Clinical outcomes differ based on esketamine combination is flawed because there is no true comparator group receiving only antidepressants alone.

- More importantly: Lack of transparency with TriNetX data
  - Composition of contributing centers, methods of data capture, and quality assurance processes remain unclear

- Reliability concerns

- Other concerns:
  - No mention of how missing data was handled
  - Sex and Age was used for propensity score matching but no other confounders were considered

Non-Reproducibility of Real-World Data Research: A Case Study

By Nichole Carlson, PhD

Professor and Interim Chair Biostatistics and Informatics

Director, Center for Innovative Design and Analysis

Director, Biostatistics, Epidemiology, and Research Design (BERD), CCTSI

Link: Webinar

CASE STUDY OF REPRODUCIBILITY

A statistics review was done on a “random” manuscript titled “Obstetric complications in mothers with ADHD.” The manuscript looks to understand the role of Attention Deficit Hyperactivity Disorder (ADHD) as a risk factor for experiencing obstetric complications

- Retrospective observational study
- Primary exposure: ADHD diagnosis (yes/no)
- Primary Outcome: 18 outcomes based on prior research on epidemiology of pregnancy complications…Nichole looked at:
  - Pre-eclampsia
  - Pre-existing hypertension complicating pregnancy, childbirth, and puerperium
  - Gestational diabetes
  - Depressive Episode
  - Cardiac disease complicating pregnancy

THE Manuscripts Analysis Plan

Using the TriNetX analytic platform:

- Identify cohort of pregnancies
- Identify the exposure (ADHD: yes or no)
- Identify the outcomes
- 1:1 propensity matching on age, race, and ethnicity
- Default Output: Absolute risk, relative risk, and odds ratios with 95% CI, t-test P value were provided by TriNetX

CLASSIFICATIONS OF REPRODUCIBILITY

- Methods rigor – do independent researchers define the same cohort and code from a research question? This requires a clearly defined research hypothesis

- Results reproducibility – do identical methods yield the same estimates (i.e. means, hazard ratios)? This requires transparent reporting and static data

- Inferential reproducibility – do researchers reach similar conclusions despite methodology or data differences?

METHODS RIGOR

Nichole Carlson investigated three approaches to selecting the pregnancy cohort:

- Manuscript cohort
- TriNetXrecommendation approach
- Team Science approach

RESULTS REPRODUCIBILITY

- Applied same approach multiple times, will we achieve the same estimates and p -values?
- We conducted each of our three cohort selection criteria 2 months apart on the platform to assess whether it matters when we access the platform
- We imposed a 5 -year outcome window (time between pregnancy and the occurrence of outcomes) to examine the influence of the restriction on the results.
  - In theory, these constraints should yield an identical cohort, enabling us to assess the impact of restricting outcome window on effect sizes, precision and overall conclusions

INFERENTIAL REPRODUCIBILITY

- Compared p -values and direction of estimation across all of the different approaches of analysis
- If we arrive at the same scientific conclusion even without methods reproducibility/rigor and results reproducibility, we could claim robustness in our findings

MANUSCRIPT COHORT

- Pregnancy from one ICD code (Z33)
- ADHD using ICD code (F90)
- First vs Any pregnancy not specified so Nichole’s team did both
- TriNetX offers a time constraint feature that allows us to specify that the recorded ADHD diagnosis must occur before pregnancy, the original paper did not specify any such constraint, so we did not apply temporal ordering of the exposure (ADHD) and pregnancy
- ICD codes for outcomes were listed in the manuscript

TriNetX Recommendations Cohort

- Best practices for querying maternal and fetal medicine is available to registered TriNetX users… it is not public access, so it is hard to find as a reviewer

- “Advanced Analytics Strategy Guide” emphasizes the importance of setting an appropriate time window to define the period during which exposure and outcomes will be observed relative to the index event.

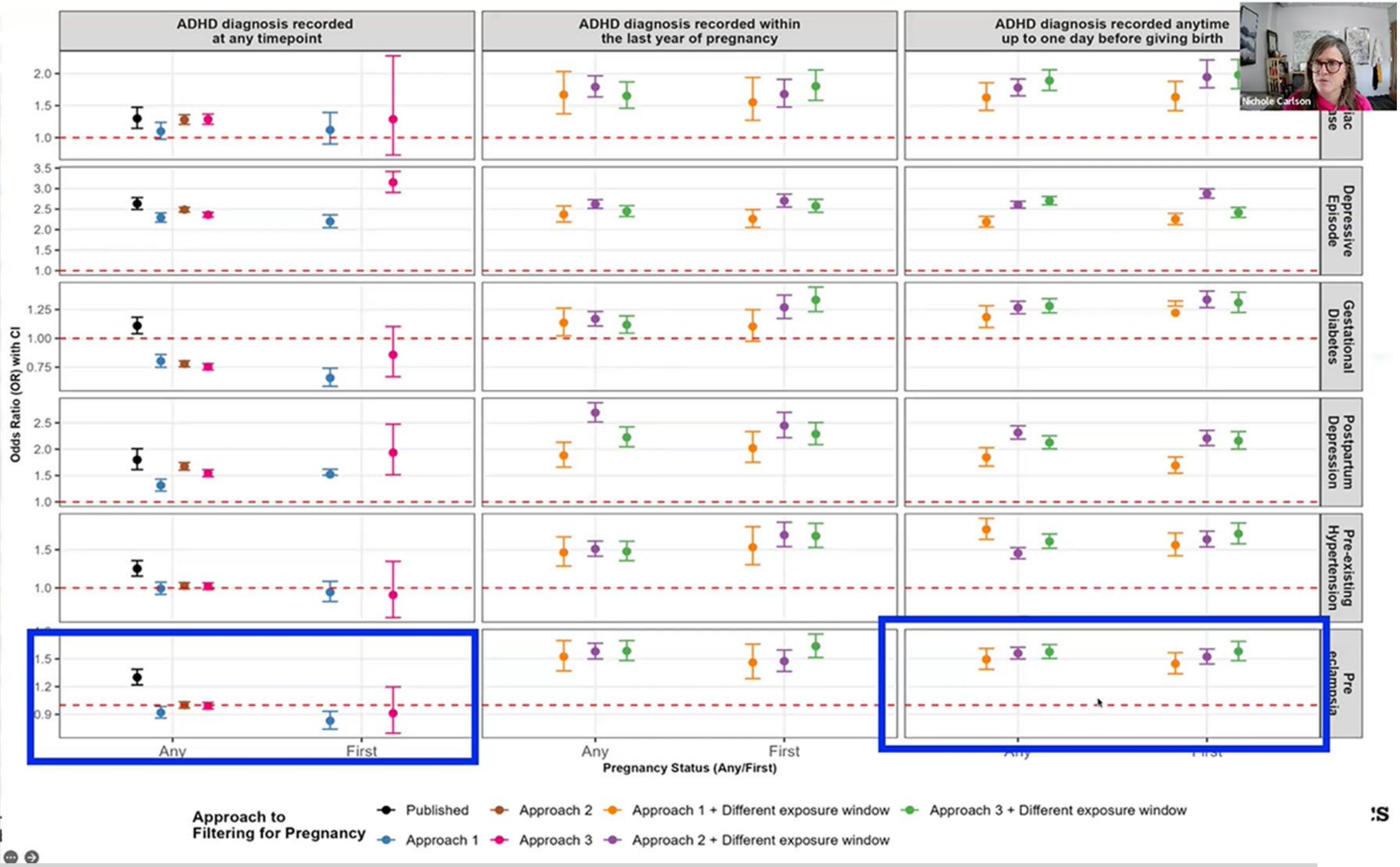
- Nichole’s team used the TriNetX Recommendations:
  - Identifying pregnant individuals using recommended expanded ICD code
  - Defining a time window between ADHD diagnosis and pregnancy
    - Any instance of ADHD occurred within the last year prior to pregnancy
    - Any instance of ADHD occurred at least one day prior to pregnancy

Team Science Cohort

- Collaborated with two expert OBGYN researchers who been building an EHR based platform for identifying pregnancy women who are receiving care at the location of their records
- ICD code list is very extensive
- Dozens and dozens of ICD code combinations and has been validated with chart reviews

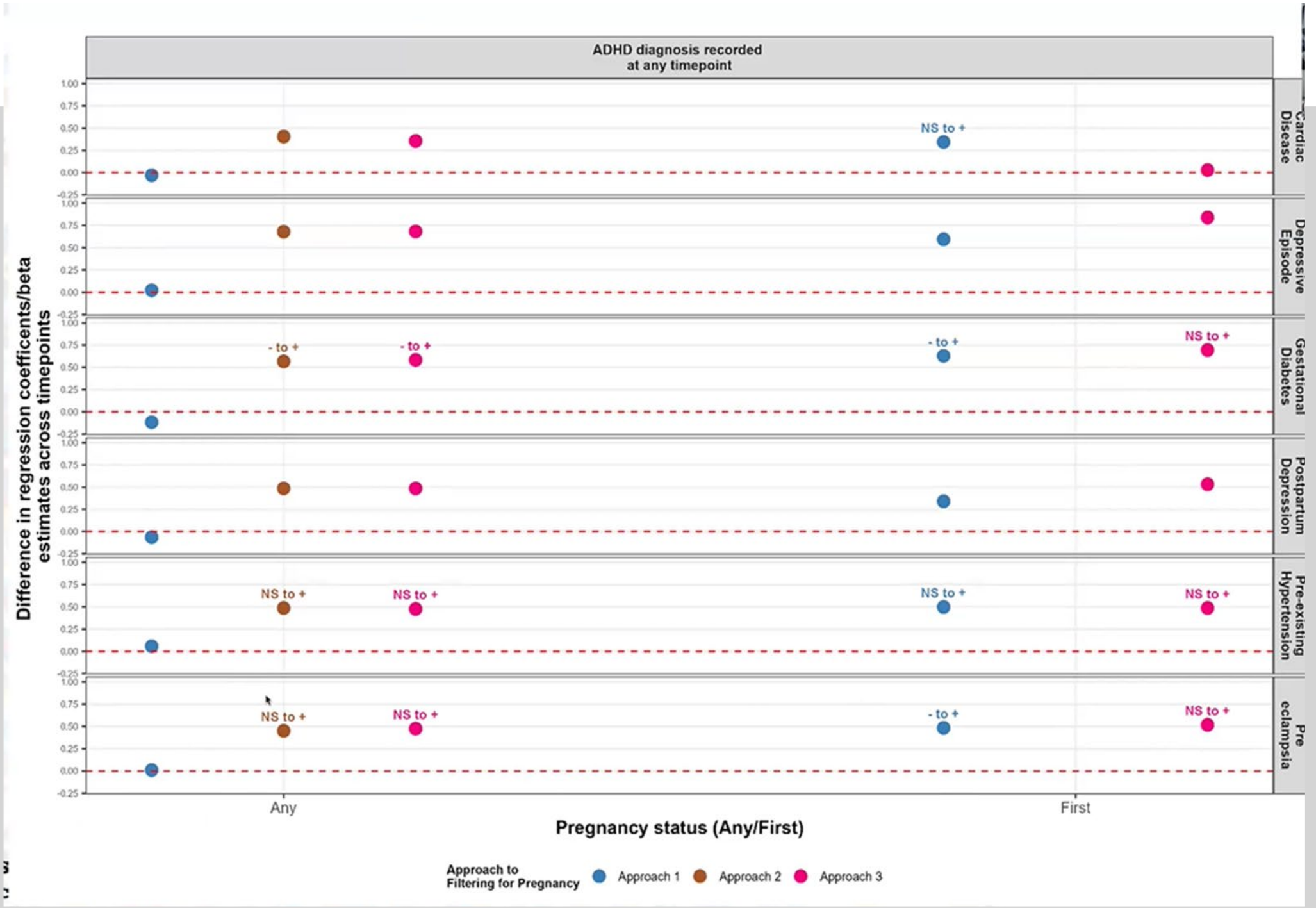
REPRODUCIBILITY

Time window between any record of ADHD diagnosis and pregnancy		Pregnancy Instance (First/Any)	Record size (RS)s		
Start of time window	End of time window		Approach 1	Approach 2	Approach 3
None (Published finding)	None (Published finding)	Unknown	45,737		
None (Following paper's methods)	None (Following paper's methods)	Any	33,108	162,189	129,435
None (Following paper's methods)	None (Following paper's methods)	First	17,223	30,341	21,532
Birth of the individual	1 day before giving birth to child	Any	20,203	83,430	57,635
Birth of the individual	1 day before giving birth to child	First	19,648	53,880	31,091
1 year before pregnancy	1 day before giving birth to child	Any	10,561	53,223	27,758
1 year before pregnancy	1 day before giving birth to child	First	7,844	24,386	21,210


ADHD diagnosis recorded at any timepoint
ADHD diagnosis recorded within the last year of pregnancy
ADHD diagnosis recorded anytime up to one day before giving birth
Odds Ratio (OR) with CI
Depressive Episode
Gestational Diabetes
Postpartum Depression
Pre-existing Hypertension
Pre eclampsia
Any
First
Pregnancy Status (Any/First)
Approach to Filtering for Pregnancy
Published
Approach 1
Approach 2
Approach 3
Approach 1 + Different exposure window
Approach 2 + Different exposure window
Approach 3 + Different exposure window

LOOKING AT RESULTS REPRODUCIBLITY

Method	Time Window between any instance of ADHD and Pregnancy indicator in data		Pregnancy Instance (First/Any)	RS at timepoint1	RS at timepoint 2	Difference
	Start of time window	End of time window				
1	None (Following paper's methods)	None (Following paper's methods)	Any	33,108	44,657	11,549
1	None (Following paper's methods)	None (Following paper's methods)	First	17,223	20,246	3,023
1	Birth	1 day Before	Any	20,203	25,922	5,719
1	Birth	1 day Before	First	19,648	22,410	2,762
1	1 Year before pregnancy	1 day before	Any	10,561	25,922	15,361
1	1 Year before pregnancy	1 day before	First	7,844	9,781	1,937


ADHD diagnosis recorded at any timepoint
Cardiac Disease
Depressive Episode
Gestational Diabetes
Postpartum Depression
Pre-existing Hypertension
Pre eclampsia
NS to +
- to +
Difference in regression coefficents/beta estimates across timepoints
Any
First
Pregnancy status (Any/First)
Approach to Filtering for Pregnancy
Approach 1
Approach 2
Approach 3

DISCUSSION

- Large proportion of manuscripts did not include a biostatistician and the tools available do not include advanced methods
- Different expertise on the team results in different cohorts
- Different questions of interest could be answered based on different methods of cohort selection
- Risks overstating finding or over generalizing findings when using RWD platforms

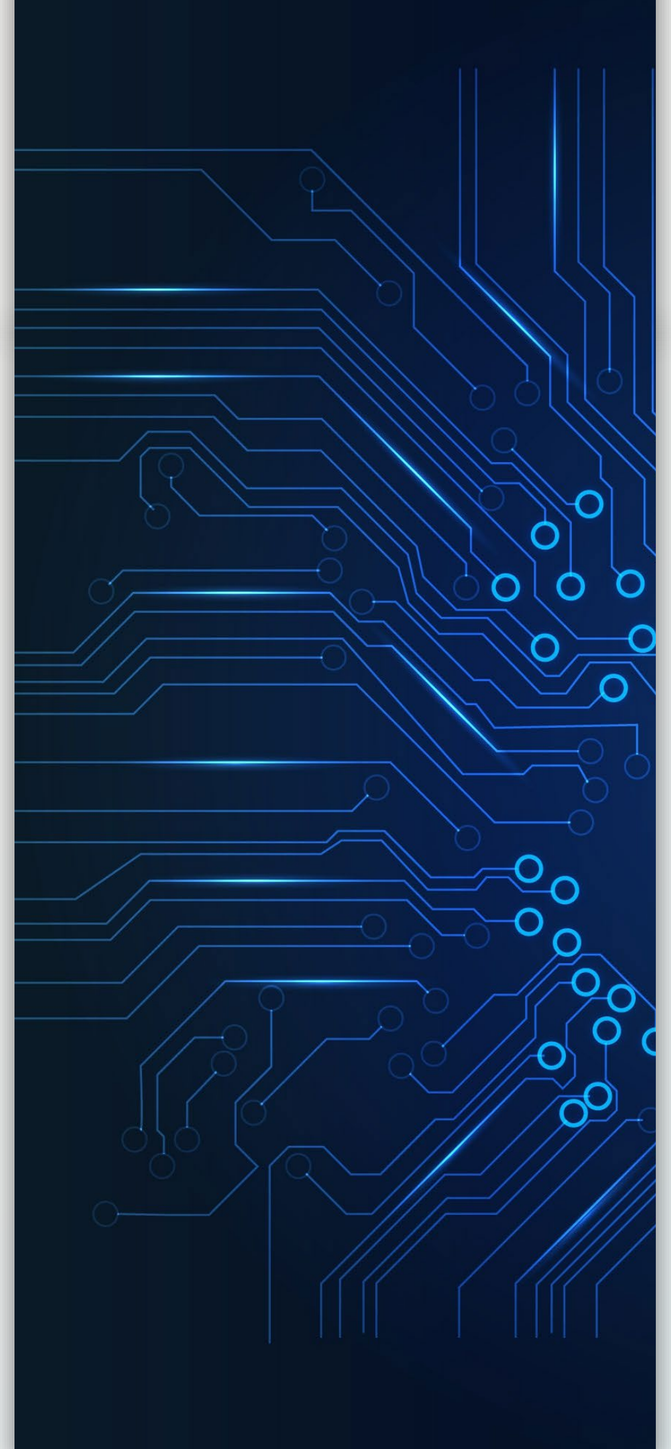
DISCUSSION

- Adhere to team science -based principles
  - Just because you can does not mean you should
  - Add a quantitative expert on the team
  - Add a data expert to the team (works with TriNetXa lot)
  - **Add a subject area expert to the team**

- Advice on:
  - Careful cohort definition
  - Careful exposure definition
  - Temporal ordering

- Association vs. Causation

- Advice on at least looking at the TriNetX Recommendations



THANK YO